
Beyond Lexical Matching: An Interpretable Analysis of Resume–Job Matching and Ranking

Shiyi Li
2026-05-08¹

Abstract

This project studies automated resume–job matching using interpretable machine learning models. Given resume–job pairs, the task is to predict hiring decisions and rank candidates by relevance. The analysis compares keyword-based methods, TF-IDF representations, and engineered alignment features such as overlap ratio, cosine similarity, and skill matching. Results show that lexical representations dominate predictive performance, while engineered features provide only limited improvements. Despite moderate classification accuracy, ranking metrics are nearly saturated across models. Further analysis indicates that this is largely driven by dataset structure, where small candidate pools simplify the ranking task. These results suggest that lexical features alone are insufficient for robust hiring systems, motivating stronger semantic representations and continued human supervision.

1. Introduction

Efficient talent matching remains a fundamental challenge in modern labor markets. In practice, identifying the most suitable candidate among a large pool of applicants is a time-consuming and resource-intensive process, often involving multiple stages such as resume screening and several rounds of interviews.

Two types of errors are particularly costly in this process. False negatives may lead to the rejection of highly qualified candidates, resulting in a loss of potential talent for the firm. On the other hand, false positives increase the workload of human resource departments by advancing unqualified candidates to later interview stages. Therefore, reducing both false positives and false negatives is essential for improving hiring efficiency.

To address this problem, we propose an automated resume–job matching system that evaluates the relevance between candidate resumes and job descriptions at the screening stage. Using historical hiring data with binary labels

(selected vs. rejected), we construct models that not only learn from textual representations but also explicitly incorporate alignment-based features. In particular, we introduce overlap ratio to capture exact lexical matching between job requirements and resume content, and cosine similarity over TF-IDF representations to model global textual alignment. .

Specifically, given a resume and a job description, our objectives are: (1) to predict whether the candidate should be selected or rejected, and (2) to rank candidates based on their predicted relevance.

We evaluate our system from both classification and ranking perspectives. For classification, we report accuracy, precision, recall, and F1-score to measure predictive performance. For ranking, we use Precision@k and NDCG to assess the quality of candidate ordering. Beyond standard evaluation, We compare multiple baselines, including keyword-based and TF-IDF-based models, to ensure that observed performance is not solely driven by simple lexical overlap. Rather than focusing on complex deep learning architectures, this work prioritizes feature engineering and interpretable machine learning approaches.

This analysis examines how resume–job alignment features affect classification and ranking performance, complemented by error analysis and learning-curve diagnostics to identify model limitations. It further reveals that ranking metrics can become artificially saturated due to small candidate pools in the dataset. Finally, the findings highlight important implications for real-world deployment, including fairness and bias concerns arising from lexical dependence and annotation artifacts.

2. Related Work

Previous studies on automated resume screening and candidate matching cover a wide range of approaches, from traditional keyword-based systems to more advanced NLP and embedding-based methods.

Many recent studies focus on semantic representation and deep learning techniques. Jaiswal et al. [1] introduce a resume analysis and job recommendation framework built on BERT and spaCy, using named entity recognition and

contextual embeddings to extract structured information from resumes. Although this approach improve semantic representation, they often reduce model interpretability.

Another research direction combines classical NLP pipelines with similarity-based matching methods. Daryani et al. [2], Gangoda et al.[3], and Saatçi et al.[4] (2024) transform resumes and job descriptions into TF-IDF representations and apply similarity measures such as cosine similarity or Jaccard similarity for candidate ranking. These studies emphasize lexical alignment and skill matching, suggesting that explicit similarity measures can work effectively for ranking tasks. However, many of these approaches rely heavily on a single similarity metric and may not fully capture the complexity of resume-job matching.

Several studies also examine traditional machine learning methods for resume classification and recommendation. Surya et al.[5] apply models including SVM, Logistic Regression, KNN, and Random Forest using TF-IDF features, showing that classical models can still achieve competitive performance with proper preprocessing. However, these studies mainly focus on classification accuracy and provide limited analysis of ranking quality and feature-level contributions.

Unlike prior work that mainly emphasizes deep semantic models, our project focuses more on interpretable baselines and feature-engineering analysis. Instead of depending on a single representation or model, we compare multiple feature groups, including TF-IDF, overlap ratio, cosine similarity, and skill-alignment features, through an ablation-style analysis. In addition, we evaluate both classification and ranking performance using metrics such as Precision@10 and NDCG@10, while also analyzing model behavior and feature contributions in detail. This helps us better understand how different alignment signals influence resume-job matching performance.

3. Dataset

The Resume Screening Dataset [6] is a structured tabular NLP dataset designed for modeling resume–job matching, supporting both classification and ranking tasks.

- **Scale and Split:** Approximately 10.2k candidate–job pairs with an 80/20 train–test split
- **Data Structure:** Each observation represents a resume–job pairing instance
- **Core Fields:**
 - *role*: categorical job title (45 unique classes)
 - *resume*: unstructured text describing candidate experience and skills

- *job description*: text outlining job requirements
- *reason for decision*: annotated natural-language explanation of hiring outcome
- *decision*: binary label (selected vs. rejected)

- **Class Distribution:** Nearly balanced ($\approx 50.3\%$ rejected, 49.7% selected), enabling standard supervised learning without reweighting
- **Modeling Implication:** Predicted probabilities can be directly extended to ranking candidates within each job description

3.1. EDA Analysis

Before comparing models, we conducted Exploratory Data Analysis (EDA) to identify the dataset’s characteristics and potential benchmark limitations. The EDA suggests that the dataset is relatively clean and lexically structured compared with noisier real-world hiring data, providing easier ranking evaluation than for realistic hiring deployment.

First, the label distribution is almost balanced, with 5,114 reject samples and 5,060 select samples, showing no severe class imbalance issue.

Second, the vocabulary and word-frequency analysis shows that the text contains many repeated high-frequency words. The combined resume and job-description vocabulary after preprocessing has 33,647 unique tokens. The resume vocabulary is much larger than the job-description vocabulary, but both selected and rejected samples share many common words. This creates a high-dimensional sparse text representation and motivates TF-IDF instead of relying only on raw BoW counts.

Table 1. Vocabulary Size Summary

Statistic	Value
Combined vocabulary size	33,647
Resume vocabulary size	20,294
Job-description vocabulary size	2,999
Dataset size	10,174

Third, the distributions of overlap ratio and cosine similarity (Figures 1 and 2) show that surface-level alignment features do not clearly separate selected and rejected applications. As shown in Figure 1, the two classes exhibit substantial overlap, and rejected applications even display a slightly higher average overlap ratio than selected ones. Similarly, Figure 2 shows that cosine similarity distributions are nearly identical across the two classes, with both concentrated in a similar range and exhibiting comparable variance.

These patterns indicate that neither exact lexical matching nor global TF-IDF–based similarity alone is sufficient to distinguish hiring outcomes. In particular, higher lexical

overlap or textual similarity does not necessarily imply better candidate–job fit, suggesting that surface-level alignment features capture only limited aspects of relevance.

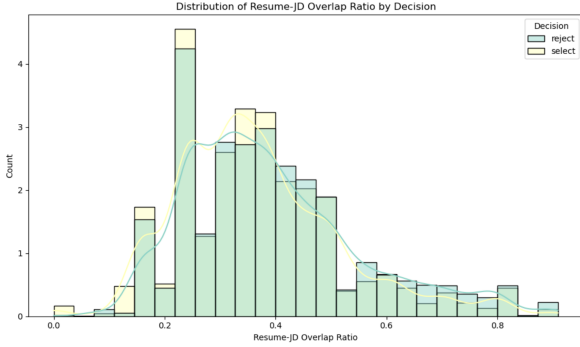


Figure 1. Distribution of resume-job overlap ratios for selected and rejected applications.

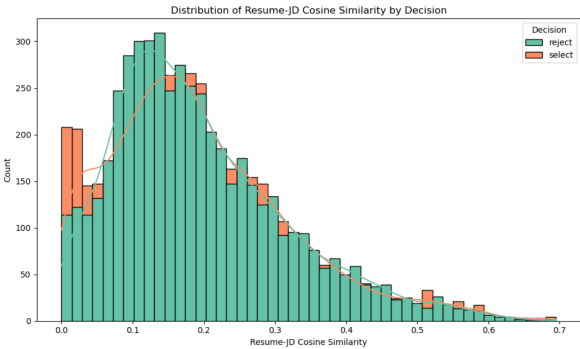


Figure 2. Distribution of TF-IDF cosine similarity between resumes and job descriptions across decision classes

Further analysis of model coefficients reveals that some highly weighted features correspond to evaluative or label-associated terms, such as *excellent*, *select*, *limited*, *gaps*, *lack*, and *reject* (Figure 3). These terms are likely tied to annotation-style language in the dataset rather than intrinsic candidate qualifications. As a result, the dataset may be partially separable through such lexical cues, which can simplify certain predictive tasks.

Taken together, these findings suggest that the dataset places stronger emphasis on lexical and evaluative wording patterns than on deeper semantic understanding of candidate qualifications. This has important implications for evaluation: while ranking metrics may appear relatively strong, classification performance remains moderate, indicating that the benchmark structure may partially simplify the ranking task without fully capturing true hiring complexity.

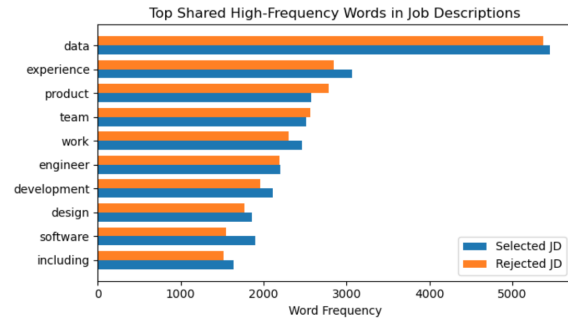


Figure 3. Top shared high-frequency words in selected and rejected job descriptions

4. Methodology

4.1. Text Representation

We compare BoWs and TF-IDF text representations. BoW is used as a simple keyword-frequency baseline. TF-IDF is used because it downweights overly common words and gives more weight to informative terms. The EDA shows that raw word frequency alone is limited because many common resume and job-description terms appear in both selected and rejected classes. Therefore, TF-IDF is a more appropriate representation for the main linear models.

4.2. Feature Engineering

The engineered features are motivated by the EDA results. Length-based features have very small differences across labels, so document length alone is not useful enough. We engineer a set of resume–job alignment features to capture complementary aspects of candidate–job matching beyond raw text representations. These features are designed to model explicit requirement coverage, global textual relevance, and domain-specific skill alignment.

- **Overlap Ratio:** We define a set-based overlap measure between the resume (R) and job description (J):

$$\text{Overlap}(R, J) = \frac{|R \cap J|}{|J|} \quad (1)$$

where R and J denote the sets of unique tokens in the resume and job description, respectively. This feature measures the proportion of job-description vocabulary covered by the resume. Using set-based overlap removes duplicate-token effects and avoids over-weighting repeated keywords, providing a normalized and interpretable measure of explicit lexical alignment.

- **Cosine Similarity:** We compute cosine similarity between TF-IDF representations of resumes and job de-

scriptions. This feature captures global textual alignment by incorporating term importance, allowing the model to measure overall relevance beyond exact token overlap.

- **Skill Match Features:** We construct explicit binary indicators for the presence of key technical skills (e.g., *Python*, *SQL*, *AWS*, *Docker*) in both the resume and job description. These features focus on domain-specific alignment and capture whether essential skills required by the job are explicitly matched by the candidate.

4.3. Experimental Setup

We use an ablation-study style experiment to measure how each feature group contributes to performance. The experiment starts from text-only baselines and then adds engineered alignment features step by step.

Table 2. Ablation Study Configuration

Model	Configuration
Baseline	Bag-of-Words + Multinomial Naive Bayes
A	TF-IDF + Logistic Regression
B	TF-IDF + Linear Support Vector Machine
C	TF-IDF + TruncatedSVD + Logistic Regression
D	TF-IDF + overlap ratio + Logistic Regression
E	TF-IDF + overlap ratio + cosine similarity + Logistic Regression
F	TF-IDF + overlap ratio + cosine similarity + skill features + Logistic Regression
G	Engineered features + Random Forest
H	Engineered features + XGBoost

5. Evaluation Metrics

We evaluate model performance using both classification and ranking metrics. For classification, we report accuracy, precision, recall, and F1-score, where F1-score is particularly important as it balances false positives and false negatives. For ranking, we use Precision@10 and NDCG@10 to assess whether the most relevant candidates appear near the top of the ranked list.

However, ranking metrics in this dataset are nearly saturated across models and thus provide limited discriminative power. This phenomenon is primarily driven by dataset characteristics rather than true model differences. Specifically, many job descriptions are associated with extremely small candidate pools: in the holdout set, a large proportion of jobs have only one or two candidates, and 88.1% of jobs contain at most one positive example. As a result, ranking tasks often reduce to identifying a single obvious positive candidate rather than distinguishing among multiple competitive candidates.

This leads to artificially high scores not only for Preci-

Table 3. Classification Performance

Model	Acc	Prec	Rec	F1
BoW + NB	.559	.539	.770	.634
TF-IDF + LR	.555	.552	.564	.558
TF-IDF + SVM	.560	.558	.549	.554
TF-IDF + SVD	.560	.555	.585	.570
+ overlap	.555	.551	.564	.558
+ cosine	.551	.547	.566	.556
+ skill	.552	.549	.564	.556
RF	.514	.511	.489	.500
XGB	.503	.500	.533	.516

sion@10 and NDCG@10, but also for stricter metrics such as Precision@1, Recall@1, MRR, and MAP. Additionally, ranking outputs across different models are highly correlated, further indicating that dataset structure dominates model performance. Therefore greater emphasis should be placed on classification metrics when comparing models.

6. Experimental Results

6.1. Classification Performance

The classification results (Table 3) show that lexical representations dominate predictive performance in this dataset. The BoW + Naive Bayes model achieves the highest F1-score (0.634), driven by a substantially higher recall (0.770), indicating strong ability to capture true positives at the cost of more false positives.

Among TF-IDF-based models, TF-IDF + Logistic Regression produces balanced precision and recall (F1 = 0.558), while TF-IDF + SVD + Logistic Regression achieves the best performance within this family (F1 = 0.570), suggesting that dimensionality reduction helps mitigate sparsity noise.

The ablation study shows that adding engineered alignment features (overlap, cosine similarity, and skill matching) results in only marginal changes. Performance remains around F1 0.55 across all variants, indicating limited incremental gains from these features.

Models trained solely on engineered features perform worse. Random Forest achieves F1 = 0.500 and XGBoost achieves F1 = 0.516, suggesting that compact alignment features alone cannot replace full text representations.

Figure 4 further shows that classification metrics are consistently clustered in a narrow range, indicating limited separability between models.

6.2. Feature Analysis

We further analyze the discriminative strength of engineered alignment features using descriptive statistics.

The overlap-ratio feature captures explicit lexical alignment

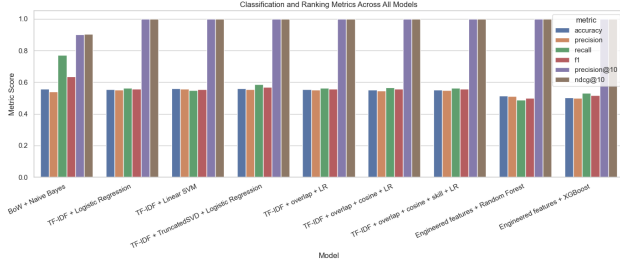


Figure 4. Classification and ranking metrics across all models

between resumes and job descriptions. However, the distributions show weak separation: rejected samples have a slightly higher mean overlap (0.392) than selected samples (0.367), with medians of 0.364 and 0.333, respectively. This counterintuitive pattern suggests that higher lexical overlap does not necessarily imply better candidate–job fit, and that exact token matching alone is insufficient to distinguish outcomes.

Cosine similarity, which measures weighted textual alignment using TF-IDF representations, also shows minimal separation. The mean cosine similarity is 0.194 for rejected samples and 0.192 for selected samples, with nearly identical medians (0.170 vs. 0.172). The distributions largely overlap, indicating that global textual similarity is not a strong standalone predictor. This may be due to the prevalence of shared professional vocabulary across both classes.

Skill-matching features provide interpretable signals of domain-specific alignment, but their overall magnitude remains small. The average skill-match rate is approximately 1.83% (median 1.53%), and even the largest differences across classes are modest. For example, technologies such as *Java*, *AWS*, *Azure*, *Python*, and *Docker* appear slightly more frequently in selected samples, but the largest observed difference is less than one percentage point.

Overall, all engineered alignment features exhibit weak standalone separation. While they capture meaningful aspects of resume–job matching, their primary contribution lies in interpretability rather than substantial improvements in classification performance.

7. Error Analysis

A central component of this project is failure-mode analysis. The main Logistic Regression model exhibits a relatively balanced confusion profile, with 470 false positives and 441 false negatives. This indicates that the classifier does not strongly favor one class, but instead makes comparable errors in both directions.

The presence of a large number of both false positives and

false negatives suggests that many resume–job pairs lie close to the decision boundary. Consequently, improving performance likely requires features that better separate these borderline cases, rather than simply adjusting the classification threshold.

Figure 5 illustrates this behavior. The confusion matrix shows that errors are distributed across both classes rather than concentrated in a single direction. This supports the interpretation that many resume–job pairs remain inherently ambiguous under lexical feature representations.

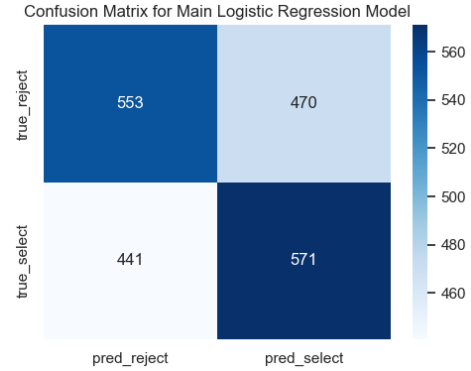


Figure 5. Confusion-matrix heatmap for the main Logistic Regression model using TF-IDF, overlap, cosine similarity, and skill feature

7.1. False Positives

False positives correspond to cases where the model predicts *select* but the true label is *reject*. These examples often have relatively high predicted probabilities (e.g., 0.719, 0.700, 0.692) and typically contain well-structured resume language with relevant-looking keywords. As a result, the model assigns them high scores despite the ground truth being negative.

This pattern suggests that keyword-heavy or well-formatted resumes can mislead lexical matching models. In other words, the model may overreact to surface-level relevance without capturing deeper job-specific fit.

7.2. False Negatives

False negatives occur when the model predicts *reject* for candidates that are actually *selected*. These cases typically have low predicted probabilities (e.g., 0.224, 0.254, 0.263).

One common reason is that relevant qualifications are expressed using different wording from the job description. This reduces overlap-based and lexical matching signals, even when the candidate is genuinely suitable. These cases highlight the limitations of exact lexical matching and motivate the need for richer semantic representations.

The prediction-score distribution further supports this interpretation. False positives and false negatives tend to cluster near the decision threshold (0.5), with average predicted probabilities of approximately 0.560 and 0.442, respectively. This indicates that most errors arise from uncertain boundary cases rather than highly confident misclassifications.

Figure 6 provides additional insight. Selected applications exhibit a heavier right tail in predicted probabilities, which allows the model to rank strong candidates near the top. At the same time, the overlap between selected and rejected distributions remains substantial, explaining the difficulty in classification.

This behavior also connects directly to the observed ranking saturation. The per-job candidate-pool analysis shows that many job descriptions have extremely small candidate sets: 905 out of 1,265 jobs have only one candidate, 204 have exactly two, and 88.1% contain at most one positive example.

As a result, many ranking tasks are effectively trivial. The model only needs to place a single clear positive candidate above a very small number of alternatives, rather than solve a complex ranking problem with many competitive candidates. This explains why ranking metrics can remain near-perfect even when the classification boundary remains noisy.

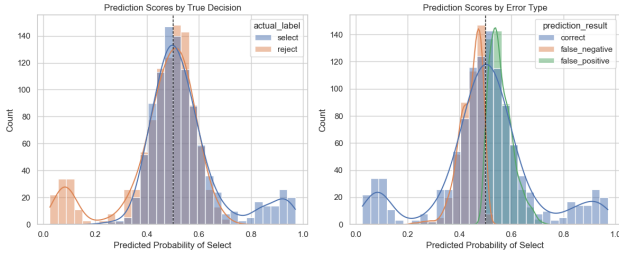


Figure 6. Prediction score distributions for the main Logistic Regression model

8. Discussions

8.1. Lexical Dominance in Classification

The results indicate that classification performance is primarily driven by lexical frequency patterns rather than structured alignment signals. The strong performance of Naive Bayes suggests that token-frequency statistics alone capture a substantial portion of the predictive signal, while engineered features such as overlap ratio, cosine similarity, and skill matching provide only marginal improvements.

This suggests that the model mainly depends on surface-level lexical patterns instead of deeper semantic matching.

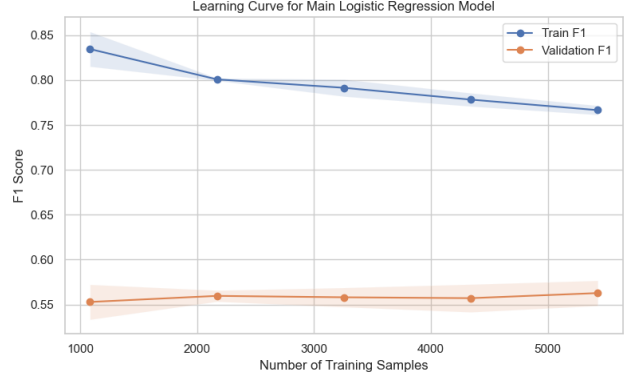


Figure 7. Learning curve for the main Logistic Regression model using TF-IDF, overlap, cosine similarity, and skill-match features.

In effect, many hiring decisions in this dataset can be approximated through word-level patterns that correlate with selection outcomes.

Consequently, the classification task appears to be governed more by lexical separability than by true semantic matching. This highlights a key limitation of the benchmark: strong predictive performance may reflect the ability to exploit word-frequency patterns rather than to capture meaningful relationships between candidate qualifications and job requirements.

8.2. Training and Learning Curve Diagnostics

Figure 7 presents the learning curves of the main Logistic Regression model in terms of F1-score across increasing training set sizes.

Two key patterns emerge. First, the training F1-score decreases steadily as the number of training samples increases, dropping from approximately 0.84 to 0.77. This behavior is expected, as the model is exposed to more diverse and difficult examples, reducing its ability to overfit the training data.

Second, the validation F1-score remains relatively stable, fluctuating slightly around 0.55–0.56, with only marginal improvement as more data is added. Notably, the gap between training and validation performance persists across all sample sizes, indicating that the model does not fully generalize to unseen data.

Together, these patterns suggest that the model operates in a regime characterized by moderate bias and limited variance reduction. The persistent gap between training and validation performance indicates that increasing the amount of training data alone is unlikely to significantly improve generalization performance. Instead, the bottleneck appears to lie in the representational capacity of the feature space.

In particular, the plateauing validation performance is consistent with earlier findings that lexical and alignment-based features provide only limited separability. While the model can fit training data reasonably well, it struggles to resolve ambiguous boundary cases in the validation set.

This diagnostic reinforces the conclusion that further performance improvements are more likely to come from richer feature representations—such as semantic embeddings or contextual models—rather than from additional data or further tuning of the current model class.

8.3. Lexical Decision Patterns in Logistic Regression

To better understand the internal decision mechanism of the main model, we analyze the largest positive and negative coefficients learned by the Logistic Regression classifier (Figure 8).

The coefficient distribution reveals a strong asymmetry between positive and negative signals. Positive coefficients are dominated by general evaluative and descriptive terms such as *organization*, *domain*, *fluent*, *known*, and *select*. These terms tend to reflect favorable or confident descriptions of candidate qualifications, rather than specific technical skills.

In contrast, negative coefficients are concentrated around words such as *limited*, *areas*, *struggled*, *lack*, *interview*, and *reject*. These terms are strongly associated with negative evaluation or assessment outcomes, suggesting that the model has learned to identify linguistic patterns that signal deficiencies or rejection.

This pattern indicates that the model relies heavily on evaluative language and annotation-style vocabulary, rather than purely on objective skill-based matching. In particular, many high-weight features correspond to subjective assessment terms that are likely introduced during the labeling process, rather than inherent candidate qualifications.

As a result, the model appears to exploit lexical shortcuts, using easily identifiable wording patterns that correlate with labels instead of learning deeper semantic relationships between resumes and job descriptions. This helps explain the strong performance of simple frequency-based models such as Naive Bayes, as well as the limited gains from more structured alignment features.

Overall, the coefficient analysis suggests that classification performance in this dataset is driven more by lexical separability and labeling artifacts than by genuine candidate–job compatibility. This highlights a key limitation of the benchmark: high predictive accuracy may reflect the ability to detect annotation-related wording patterns rather than true hiring relevance.

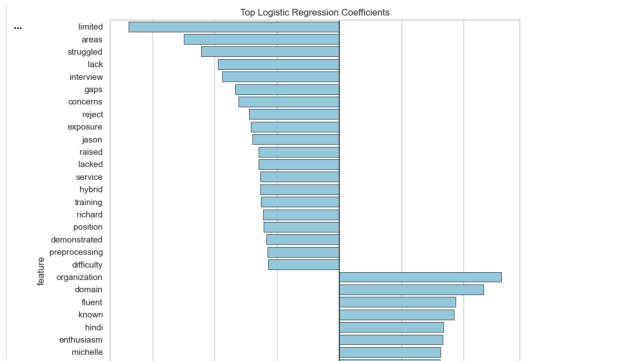


Figure 8. Top positive and negative Logistic Regression coefficients learned from TF-IDF features

8.4. Feature Behavior Analysis

We further analyze how different models utilize engineered alignment features by comparing feature-importance distributions in Random Forest and XGBoost (Table 4).

The two models exhibit distinct patterns. Random Forest assigns a dominant share of importance to cosine similarity, indicating that it relies heavily on global textual alignment as the primary predictive signal. In contrast, XGBoost distributes importance more evenly across cosine similarity, overlap ratio, and skill-based features. This suggests that boosted tree models are better able to exploit complementary interactions among multiple alignment features.

Across both models, cosine similarity consistently emerges as the strongest individual feature, confirming that pairwise textual alignment provides the most informative signal among the engineered features. However, the increased importance of skill-related features in XGBoost indicates that nonlinear models can partially capture additional structure beyond simple similarity measures.

Despite these differences in feature utilization, both models remain constrained by the limited expressive power of compact engineered features. Even when features receive high importance scores, their standalone predictive strength remains insufficient compared to full text representations. This is reflected in the overall performance gap between compact-feature models and TF-IDF-based models.

Taken together, these results highlight an important distinction between *feature importance* and *feature sufficiency*. High importance indicates that a feature is useful within a given representation space, but does not imply that the feature set itself is rich enough to support strong predictive performance. In this dataset, engineered alignment features provide complementary signals, but the dominant predictive power remains driven by high-dimensional lexical represen-

Table 4. Feature Importance Comparison: Random Forest vs. XGBoost

Feature	Random Forest	XGBoost
Cosine similarity	0.816	0.326
Overlap ratio	0.163	0.334
Skill match score	0.010	0.187
Skill match ratio	0.011	0.154

tations.

9. Implications and Ethical Considerations

The results highlight important considerations for the deployment of AI-assisted hiring systems. The heavy dependence on lexical and evaluative wording patterns indicates that models may rely on dataset-specific language cues instead of broader indicators of candidate quality. This issue can become more noticeable because the dataset is relatively clean and structured, allowing machine learning models to overfit to selection-related wording and annotation patterns. In addition, variations in domain-specific vocabulary across industries may further weaken the reliability of overlap-based matching methods.

These findings also raise concerns regarding fairness and robustness. Candidates who use different wording styles, unconventional resume formats, or nonstandard expressions may be unfairly disadvantaged, while resumes containing many keywords may receive disproportionately high scores. More importantly, hiring remains a human-centered process, and reducing candidate qualifications to token-level patterns may oversimplify personal experience and professional ability.

In practice, AI screening systems should assist recruiters rather than replace human decision-making entirely. Human oversight is still necessary to support fair evaluation and reduce biases caused by dataset artifacts and limitations in text representation.

10. Limitations

This study has several limitations. First, the feature representation relies heavily on lexical and alignment-based signals, which may fail to capture deeper semantic relationships between resumes and job descriptions.

Second, the dataset itself appears to be relatively clean and lexically separable compared to real-world hiring data. In particular, the presence of annotation-style language in the `reason_for_decision` field may introduce label-associated wording patterns, allowing models to exploit superficial cues rather than true candidate–job fit.

Third, ranking evaluation is affected by the small size of

candidate pools. Many job descriptions are associated with only one or two candidates, which simplifies the ranking task and leads to near-saturation of metrics such as Precision@10 and NDCG@10. As a result, ranking performance may overstate the effectiveness of the models.

Finally, this work does not include a comprehensive fairness evaluation across demographic or linguistic groups. The analysis is limited to English-language resumes and job descriptions, which may restrict generalizability to broader hiring contexts.

11. Conclusion

Our experiments show that interpretable machine learning models are capable of handling resume–job matching tasks, while still facing important limitations.

From a modeling standpoint, lexical representations remain the main factor influencing predictive performance. Simple models such as Naive Bayes obtain strong results by capturing token-frequency patterns, whereas TF-IDF-based Logistic Regression provides a more balanced and interpretable approach. Engineered alignment features, including overlap ratio, cosine similarity, and skill matching, provide additional insights but lead to only modest improvements in classification performance.

From an evaluation standpoint, one important finding is the gap between classification and ranking metrics. Although classification performance stays moderate, ranking metrics are close to saturation across most models. Further analysis suggests that this is mainly caused by the dataset structure, where small candidate pools reduce ranking into a simplified top-candidate identification task.

Overall, these findings indicate that current performance depends more on lexical separability and dataset-specific artifacts than on genuine semantic understanding of candidate–job compatibility. As a result, improving real-world hiring systems will require richer semantic representations, more realistic evaluation settings, and careful incorporation of human judgment.

Ultimately, resume–job matching should be viewed as a decision-support task rather than a fully automated prediction system, where interpretability, fairness, and robustness are considered just as important as predictive accuracy.

- [1] G. Jaiswal, A. Uttam, D. D. Dubey, and P. K. Mall, “Resume Analyser and Job Recommendation System Based on NLP,” in 2024 2nd International Conference on Disruptive Technologies (ICDT), 2024.
- [2] C. Daryani, G. Singh Chhabra, H. Patel, I. Kaur Chhabra, and R. Patel, “An Automated Resume Screening System Using Natural Language Processing and Similarity,” *Ethics and Information Technology (ETIT)*, vol. 2, no. 2, pp. 99–103, 2020.
- [3] N. Gangoda, K. P. Yasntha, C. Sewwandi, N. Induvara, S. Thelijjagoda, and N. Giguruwa, “Resume Ranker: AI-Based Skill Analysis and Skill Matching System,” 2024.
- [4] M. Saatçi, R. Kaya, and R. Ünlü, “Resume Screening with Natural Language Processing (NLP),” *Alphanumeric Journal*, vol. 12, no. 2, 2024.
- [5] C. H. Sai Surya, G. L. V. S. Kiriti, K. Satya Saketh, P. Sri Charan, and A. Anjali, “Optimizing Job Matching Through Automated Resume Screening with NLP and Machine Learning,” 2024.
- [6] AzharAli05.
Resume-Screening-Dataset.
Hugging Face Datasets, 2024.
Available:
<https://huggingface.co/datasets/AzharAli05/Resume-Screening-Dataset>